Report on Data Preprocessing and Preparation for ML Modeling

1. Missing Value Strategies Used and Their Effectiveness

Missing values were identified using Pandas isnull().sum() during the initial dataset exploration. Several missing-value handling techniques were implemented and compared.

For numerical variables, median imputation using SimpleImputer(strategy="median") was applied. Median imputation is useful because it is less affected by extreme values than mean imputation.

For categorical variables, most-frequent imputation was applied using SimpleImputer(strategy="most_frequent"). The gender column was also tested separately using its mode.

Additional techniques were implemented, including missing indicators with random sample imputation, KNN Imputation, MICE (Iterative Imputer), and Complete Case Analysis.

The KNN and MICE methods used relationships between numerical variables to estimate missing values, while complete-case analysis removed rows containing missing values. Overall, median and categorical-mode imputation provide a simple way to retain the dataset while avoiding unnecessary row removal.

2. Outlier Handling Results

Several techniques were used to identify and handle outliers.

The Z-score method was applied to annual_income, with observations having an absolute Z-score greater than 3 identified as potential outliers.

The IQR method was applied to loan_amount. The first quartile (Q1), third quartile (Q3), IQR, lower limit, and upper limit were calculated to identify unusually high or low values.

The percentile method was also tested on annual_income, using the 1st and 99th percentiles as limits.

Finally, Winsorization was implemented with 5% limits on both sides. This reduces the influence of extreme observations without completely removing them.

These methods helped identify extreme values and provided different approaches for making the dataset more suitable for machine-learning algorithms.

3. Encoding Methods Applied to Categorical and Numerical Variables

Different encoding techniques were applied according to the type of variable.

Categorical Variables

- Ordinal Encoding was applied to education_level because its categories have a natural order: Primary → Secondary → Graduate → Post-Graduate.
- Label Encoding was applied to the binary gender feature.
- One-Hot Encoding was applied to categorical variables such as region and loan-purpose-related variables. This converted categories into separate binary columns.

Numerical Variables

Several numerical transformation techniques were also tested:

- Binning divided income into Low, Medium, and High groups.
- Binarization converted a numerical variable into 0/1 based on a threshold.
- Quantile Binning divided numerical values into four groups: Q1, Q2, Q3, and Q4.
- K-Means Binning grouped income into three clusters using K-Means clustering.

These techniques converted variables into forms that can be more suitable for different machine-learning models.

4. Scaling and Transformations Applied and Why

Different scaling methods were implemented to bring numerical variables to appropriate scales. The project tested:

- Standardization (Z-score scaling)
- Normalization
- Min-Max Scaling
- MaxAbs Scaling
- Robust Scaling

Scaling is important because numerical variables such as income and loan amount can have very different ranges. Without scaling, some machine-learning algorithms may give greater importance to variables simply because they have larger numerical values.

Several transformations were also applied:

- Log transformation
- Reciprocal transformation
- Square-root transformation
- Box-Cox transformation
- Yeo-Johnson transformation

These transformations can help reduce skewness and make numerical distributions more suitable for machine-learning models.

A ColumnTransformer was also implemented to apply different preprocessing techniques to numerical and categorical features simultaneously.

5. Newly Engineered Features and Their Usefulness

New features were created to provide additional information from existing variables.

Date-based Features

From transaction_date, the following features were extracted:

- transaction_year
- transaction_month
- transaction_day
- transaction_weekday

These features can help machine-learning models identify time-based patterns.

Debt-to-Income Ratio

A new feature called debt_to_income_ratio was created:

Debt-to-Income Ratio = Loan Amount / Annual Income

This feature provides a relationship between a customer's loan amount and annual income and can be useful for analyzing financial risk.

Average Monthly Value

An average monthly feature was also created by dividing the selected numerical value by 6.

These engineered features add meaningful information that is not directly available from the original variables.

6. Final Dataset Shape and Readiness for ML Modeling

The project combined data from multiple sources, including:

- CSV transaction data
- Customer metadata from JSON
- Loan repayment data from SQL
- Economic indicator data

The datasets were merged using customer_id, and economic information was merged using year.

The notebook then performed missing-value handling, outlier analysis, categorical encoding, numerical transformations, scaling, and feature engineering.

A ColumnTransformer was also created to prepare numerical and categorical variables together. The notebook prints both the original dataset shape and transformed dataset shape after applying the

ColumnTransformer.

Therefore, after preprocessing, the dataset is substantially more suitable for machine-learning modeling because missing values have been addressed, categorical variables have been converted into numerical representations, numerical features can be scaled, and additional meaningful features have been created.

Conclusion

Overall, the preprocessing workflow prepared the financial dataset for machine-learning applications by addressing data quality issues, identifying outliers, encoding categorical variables, scaling numerical features, transforming distributions, and creating useful financial and time-based features. The resulting processed dataset provides a stronger foundation for building and evaluating machine-learning models.